

Perceived Labor Market Discrimination and Job Search among U.S. First- and Second-Generation Immigrants

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August 27, 2026

Abstract

This paper investigates the relationship between perceived discrimination and job search among first- and second-generation immigrants. We elicit beliefs regarding two distinct domains of discrimination: the “current workplace” and the “job search process”. We also conduct an information provision experiment to evaluate if population-level earnings specific to one’s occupation shift individual beliefs and job search behavior. We first document that immigrants perceive substantial disparities between themselves and a counterfactual native in both domains. In addition, immigrants overestimate the native–immigrant earnings gap in their occupation. Providing population-level earnings statistics successfully shifts posterior beliefs and reduces over-pessimism. In a five-month follow-up, we find that higher perceived discrimination predicts increased search probabilities and efforts. Using initial population belief errors and treatment assignment as instruments, we find that a one-standard-deviation increase in perceived discrimination raises the probability of preparatory job search by 30.2 percentage points. These results suggest that immigrants respond to perceived discrimination in their current work or in the job search process by intensifying their search effort, rather than becoming discouraged.

JEL Codes: C93, D87, F22, F24, G2, G53, O12

Keywords: labor markets, immigration, discrimination, field experiments.

1 Introduction

Immigration continues to be a widely debated issue. Much of the discussion in economics has been about the potential impact on local labor markets, especially native-born workers and firms, and the innovation process.¹ While there is much work on labor market discrimination, it is often difficult to focus on immigrants because of missing micro-level data.²

In this paper, we conduct an information provision experiment (Haaland et al., 2023) to elicit perceived discrimination by immigrants in the U.S. and assess how these beliefs affect job search. The experiment provides immigrant workers with the average salary in their occupation for two types of workers: (1) U.S.-born workers whose parents are not immigrants (we call these “native-born” workers) and (2) a worker who is of the same immigrant group as them. This can either be a first-generation immigrant (someone born outside of the U.S.) or a second-generation immigrant (someone born in the U.S. but for whom at least one parent was born outside of the U.S.), also known as, the child of an immigrant. The variation around immigrant status is potentially relevant based on prior work (e.g., Algan et al., 2010; Abramitzky and Boustan, 2022).

We document four main findings. First, while they do so to a lesser extent than first-generation immigrants, second-generation immigrants also perceive discrimination in the U.S. labor market. In other words, both immigrants and children of immigrants believe that an otherwise identical native-born worker with no immigrant parents would fare better than them in the current workplace and during job search. Second, immigrants hold pessimistic beliefs compared to the truth about the average earnings within their own occupation. First-generation immigrants overestimate the native-immigrant earnings gap by as much as 14 percentage points and second-generation immigrants do so by 5 percentage points. Third, and importantly from a policy standpoint, information provision shifts perceived discrimination about one’s labor market situation. Workers who hold more pessimistic beliefs about the true native-immigrant earnings gap, revise this perceived discrimination downward. Fourth, perceived discrimination is strongly and positively associated with job search effort, suggesting that immigrants do not take on a “defeatist” attitude. Using the information intervention as an instrument, we find some evidence that belief changes translate into job search effort five months later.

Our paper contributes to three literatures. XXAM: NEED TO IMPROVE THIS PART. CURRENTLY TOO BRIEF AND NOT SPECIFIC ENOUGH RE: OUR SPECIFIC CONTRIBUTIONS. First, we provide novel data on the perceptions and experiences of U.S. immigrants to the literature on assimilation (e.g., Aeberhardt et al., 2010; Algan et al., 2010). Our findings indicate that immigrant perceptions tend to be pessimistic relative to actual labor market outcomes based on Census data, and also conflict with some of the existing evidence. For instance, Abramitzky and Boustan (2022) report that children of immigrants tend to outperform children of native-born individuals on key economic outcomes.

Second, we add to a vast literature on discrimination (e.g., Bertrand and Duflo, 2017;

¹For example, Borjas (1989), Ottaviano and Peri (2012), Abramitzky and Boustan (2022), Bernstein et al. (2022), Doran et al. (2022), Borjas (2026), and Abramitzky et al. (2026).

²For example, Candelo et al. (2017), Neumark (2018), Collins and Zimran (2023), Cortés et al. (2023), Mason (2023), Schumacher et al. (2023), Coffman et al. (2024), Gagnon et al. (2025), Akee and Jones (2026), and Alfonsi et al. (2026).

Darity, 2022; Chelwa et al., 2022; Bohren et al., 2025), particularly in labor markets (e.g., Neumark, 2018; Mason, 2023). In so doing, we contribute to a small but growing set of randomized controlled trials to study how people from disadvantaged groups respond to discrimination (e.g., Alsan et al., 2021; Francis et al., 2026). Our findings suggest that immigrants respond strategically to perceived discrimination by exerting additional effort.

Third, we contribute to a vast literature on belief elicitation and information provision (e.g., Danz et al., 2022; Haaland et al., 2023; Stantcheva, 2023; Haaland et al., 2025), particularly in labor and education contexts (e.g., Wiswall and Zafar, 2015; Chakravorty et al., 2024; Coffman et al., 2024; Cortés et al., 2023; Arcidiacono et al., 2020, 2025). Our primary contribution in this vein is that the type of sample – immigrants – can also be impacted by a light-touch informational nudge, possibly impacting longer run behavior. XXAM: ARE WE SURE THAT THIS IS A NEW FINDING FOR IMMIGRANTS IN THIS LITERATURE Seems not. Barsbai et al. (2024) provides information for new Filipino immigrants before their departure. Batista et al. (2026) provides practical information relevant for immigrant integration, like strategies to search for jobs. (e.g., Belot et al., 2019; Coffman et al., 2024).

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Our paper contributes to three literatures. First, we add to a vast literature on discrimination (e.g., Bertrand and Dufló, 2017; Darity, 2022; Chelwa et al., 2022; Bohren et al., 2025; Francis et al., 2026), particularly in labor markets (e.g., Neumark, 2018; Mason, 2023). Most of this work measures discrimination on the employer side (e.g., Bertrand and Mullainathan, 2004; Oreopoulos, 2011). We instead fit into a thinner but growing literature on the worker side, asking how workers perceive labor market discrimination and how these beliefs about discrimination shape behavior (Angeli et al., 2023; Ruebeck, 2025; Gagnon et al., 2025; Lepage et al., 2025).³ We contribute in three dimensions. First, we document the perception of labor market discrimination among immigrants in the United States, and we elicit and clarify it separately along two dimensions following the literature: discrimination expected at the hiring stage (Angeli et al., 2023; Haaland and Roth, 2023) and discrimination expected within the workplace (Glover et al., 2017; Ruebeck, 2025; Gagnon et al., 2025). The distinction matters because the two can carry different behavioral predictions. Second, we introduce a measure that elicits immigrants’, as well as their children’s, beliefs about a variety of their own labor market outcomes alongside their beliefs about the outcomes of an otherwise identical native, so that the difference is a subjective ex-ante treatment effect of nativity. This adapts the elicitation of beliefs over counterfactual states (Wiswall and Zafar, 2015; Arcidiacono et al., 2020; Fuster and Zafar, 2023) to the discrimination setting.⁴ Third, we study how members of disadvantaged groups respond to perceived discrimination over a short- to medium-run horizon, rather than to a single application decision or a within-session effort choice.⁵ We link our measure of perceived discrimination to labor market behavior over the subsequent five months, simultaneously exploiting our information intervention experiment for identification.

³This is also broadly related to emerging literature on the role of job seekers’ beliefs and search behavior (e.g., Conlon et al., 2018; Mueller et al., 2021; Kiss et al., 2023; Jäger et al., 2024).

⁴This is similar to Lepage et al. (2025) but they focus on the response in college major choice.

⁵Related lab experiments examine responsiveness within a single session or application episode, which is identified through blind versus non-blind evaluation (Ruebeck, 2025), address visibility (Angeli et al., 2023), or employ explicit discriminatory wage schemes (Gagnon et al., 2025).

Second, we contribute to the literature on belief elicitation and information provision (e.g., Danz et al., 2022; Haaland et al., 2023; Stantcheva, 2023; Haaland et al., 2025), particularly in labor and education contexts (e.g., Wiswall and Zafar, 2015; Chakravorty et al., 2024; Coffman et al., 2024; Cortés et al., 2023; Arcidiacono et al., 2020, 2025). Our design follows Wiswall and Zafar (2015) in eliciting beliefs at two levels—about population aggregates and about the respondent’s own outcomes—and in using the error in the former as an experimental shock to the latter. We depart from that literature in three respects. First, the counterfactuals elicited in that literature are choice-specific, such as the wage a respondent expects in a major not chosen, whereas ours is identity-specific. We ask about an alternative nativity, holding the respondent’s job, employer, and productive characteristics fixed. The elicited difference is therefore a subjective ex-ante treatment effect of an ascriptive characteristic rather than of a decision, which matches the notion of discrimination we study. Second, and relatedly, our information provision experiment does not serve only as an instrument that shocks beliefs; it also asks whether perceived discrimination can be shaped by group-level aggregates. We show that these beliefs respond to a light-touch informational nudge, an objective population-level immigrant–native wage gap, indicating that beliefs about discrimination are connected with perceptions of aggregate group disparities rather than held independently of them. This complements Haaland and Roth (2023), who correct beliefs about hiring discrimination among the general population and measure policy preferences. Third, the response to the intervention crosses different dimensions. Although the signal concerns only annual earnings, posteriors shift for promotion probability and for the number of applications required to obtain a callback, suggesting that respondents update a single latent assessment of disadvantage rather than separate domain-specific beliefs.

Third, we provide the subjective counterpart to a literature on immigrant assimilation in the United States that has focused extensively on objective immigrant–native earnings and employment profiles (Chiswick, 1978; Borjas, 1985; Algan et al., 2010; Hermansen et al., 2025). Yet, what immigrants themselves expect, and how they attribute any expected shortfall, has not been measured. We elicit their beliefs about the average earnings of immigrants working in similar occupations like them, and benchmark the beliefs against the objective statistics constructed from the ACS. The wedge is starkest among the children of immigrants. Both the ACS and existing research (e.g., Abramitzky et al., 2021) place them at or above the children of native-born parents on key economic outcomes, yet the second generation in our sample underestimates its own earnings relative to the native born and anticipates discrimination against itself in hiring and in the workplace. Accurate aggregate outcomes and pessimistic individual beliefs coexist in the same population, and the source of that departure warrants further research.

In the remainder of the paper, we cover the study design (Section 2), discuss the results (Section 3), before concluding in Section 4.

2 Study Design

The study was administered through the National Opinion Research Center (NORC) AmeriSpeak panel, a probability-based panel designed to be representative of the U.S. household population. The sampled population comprised employed first- and second-generation U.S.

residents, ages 18 and older. The study consisted of three waves, each 4-5 months apart. The information provision experiment was conducted in wave 2; the job search outcomes were collected in wave 3; and prior and posterior beliefs were collected in waves 1 and 2 (additional details will be discussed further below). Wave 1 contained a pilot experiment, so subjects were re-randomized prior to the start of wave 2 to mitigate any risk of carry-over.⁶ Table 5 in the online appendix confirms that wave 1 beliefs are balanced across treatment assignment in wave 2. Overall, it took participants 18 minutes to complete all tasks (surveys and experiments) in all waves. They were paid at an approximate rate of \$30 per hour. XXAM: GIVE THE EXACT FIGURE [ZM: What figure are you referring to here, Arnaud? The payment scheme? Or others?] Figure 2 in the online appendix shows the sample size for each wave. Attrition is in the order of 16-18 percent and not differential by treatment status.

2.1 Information Provision Experiment

In the information provision experiment, respondents were randomly assigned to a treatment or a control group. The treatment group saw a figure displaying the average annual salary for their occupation based on U.S. Census data. Salary information was provided for two groups of workers: native-born workers and workers who have the same immigrant status as the respondent, i.e., either first- or second-generation workers. The information was provided as a bar chart and explained in words below the chart. It also underscored the difference in average annual salaries in absolute terms and as a percentage. Figures 3 and 4 in the online appendix represent the figures that a respondent saw if they worked in sales and were first- or second-generation respectively. The figures thus varied depending on the respondent’s occupation and immigrant status. After seeing the figure, respondents had to complete a comprehension check. If they provided an incorrect answer, they had to review the information again.

The control group received no information and instead answered unrelated demographic questions. As Table 5 in the online appendix shows, the treatment and control groups are balanced on observable characteristics at baseline thus suggesting that the randomization was successful. The verbatim questionnaires – including the experiment protocol, timeline, and flow of information – are available at [this link](#).

2.2 Beliefs

We elicited two classes of beliefs. The first class, Z_k , is about average earnings in the respondent’s occupation and was elicited prior to the intervention. Beliefs were elicited for two types of workers, k , native-born workers and workers of the same immigrant group as the respondent, that is, first- or second-generation workers respectively.

The exact framing of the question was as follows (W2PRE_POP3 on page 6 of the wave 2 questions at [this link](#)):

⁶The pilot experiment provided information on percent employed (including full-time), average annual salary irrespective of occupation, and average number of hours worked in a week for native-born, first-generation, and second-generation workers.

We want to ask what you think about the typical annual salary for two groups of working-aged adults with the same job as you:

1. Group One: People who were born in the United States and whose parents were also born in the United States, also known as, "native born" Americans.
2. Group Two: People who were not born in the United States, also known as, "immigrant" Americans [if the respondent was first-generation]

OR

People who were born in the United States and for whom at least one parent was born outside of the United States, also known as, "second generation" Americans [if the respondent was second-generation].

These beliefs enable us to measure the extent to which individuals have accurate expectations about the population-level earnings gap. It follows that these beliefs give a measure of the size of the information shock that a treated individual receives. The gap between this prior belief and the truth, therefore, serves as the instrument we exploit in the analysis in Section 3.

The second class of beliefs, X_j , represents the difference between self-beliefs about one's own labor market outcome (one set of questions) and one's beliefs about that same outcome for an otherwise identical native-born worker (a different set of questions). j indexes the outcome over which such beliefs were elicited, specifically, current wage, current hours, promotion probability, future wage, job arrival, or job search wage. These beliefs cover two distinct domains of perceived discrimination, namely the current workplace and the job search process. The beliefs were elicited both before and after the intervention, denoted X_j^{pre} and X_j^{post} respectively.

The exact framing of the questions for self-beliefs was as follows (part 2 on page 3 of the wave 2 questions at [this link](#)):

Now, we want to ask how you think things will be if you stay with your current employer in the future.

Q1. What do you think is the chance of getting promoted by your current employer within 1 year?

Q2. What do you think your annual salary will be in the next 5 years, if you stay with your current employer? By "annual salary," we mean "pre-tax annual salary."

Q3. Now, we want to ask how you think things will be if you want to find a new job. Suppose you want to find a similar job to what you have now but with a new employer. How many applications do you think you will need to send out before getting a callback for an interview?

Q4. Suppose you actively search for new jobs in the next year, what would you expect the average annual salary to be among all the job offers you got?

In the case of the otherwise identical counterfactual native, the questions were framed similarly except that the preamble stated the following (part 3 on pages 4-5 of the wave 2 questions at [this link](#)):

Consider the following hypothetical situation. Someone is just the same as you in terms of their gender, race and ethnicity, education level, work experience, ability to speak English, and ability to work EXCEPT that this person and this person’s parents are born in the U.S. This person has the same job with the same employer as you have now.

Please try your best to tell us what you think this person’s likely job situation would be. There are no right or wrong answers.

Q1. What would this person’s annual salary be?

Q2. How many hours would this person typically work in a week?

Q3-Q6. Similar to self-beliefs stated previously.

Theoretically, both perceived discrimination in the current workplace and during the job search process have ambiguous implications for job search intensity. Consider beliefs about the current workplace. If immigrants believe this penalty is universal across the labor market, the effect on their job search might be minimal, as they expect similar unfair treatment in any future role. Conversely, if they view this discrimination as tied to their specific employer, the resulting sense of unfairness and decreased work morale could erode employer attachment, thereby increasing job turnover and motivating them to search for new jobs (e.g., [Daldy et al., 2013](#); [Gagnon et al., 2025](#); [Ruebeck, 2025](#)).

The implications of job search discrimination are also ambiguous. Intuitively, higher perceived discrimination lowers the expected job finding rate, which diminishes the expected return of a marginal application and could discourage job seekers, leading them to submit fewer applications (e.g., [Angeli et al., 2023](#)). However, if an individual has a specific target for callbacks, a decrease in the expected callback rate per application might mechanically compel them to increase their volume of costly applications to secure “enough” callbacks. XXAM: WOULD BE GOOD TO ADD A FEW OTHER REFERENCES (INCLUDING IN FOOTNOTE 4), AND EXPAND A BIT. IN THE END THIS IS A KEY POINT OF OUR PAPER ⁷ Taken together, the ultimate impact of perceived discrimination on job search behavior remains theoretically ambiguous.

The implications of job search discrimination are also ambiguous, and this ambiguity is central to our paper. As discussed in [Angeli et al. \(2023\)](#), shifting perceived job-search discrimination, or the expected success rate, can have a non-monotonic effect on application effort. A decrease in the expected success rate—the callback probability in our survey—generates a substitution effect: it diminishes the expected return to a marginal application and can discourage job seekers, leading them to submit fewer applications. This serves as the mechanism emphasized in work on how blind hiring processes, or clarifying the bar for a job, encourage applications from disadvantaged groups (e.g., [Angeli et al., 2023](#); [Coffman et al., 2024](#); [Boring et al., 2025](#)). Nevertheless, perceived discrimination can also lead to an income effect, i.e., a pessimistic job seeker must exert more effort or increase their volume of costly applications in order to reach a target ([Abebe et al., 2025](#)). That target may be a

⁷This mechanism is loosely related to income targeting behavior in the labor supply literature, where researchers argue that a higher marginal return (wage) might lead to reduced labor supply if decision-makers aim to reach a specific income reference point.

given number of callbacks, a given number of interview opportunities, or an offer.⁸ In fact, some sociology literature, Pager and Pedulla (2015), finds that racial minorities cast a wider net in their job search than comparable whites, and argues that this broad search strategy is an adaptation to labor market discrimination. Taken together, the ultimate impact of perceived discrimination on job search behavior remains theoretically ambiguous.

In our setting, the income effect is plausibly the more relevant force for two reasons. The first reason is the horizon. Rather than observing a single application decision within one experiment or one hiring process, we elicit job search behavior over five months. In a one-shot application decision, individuals place greater weight on comparing the marginal benefits and costs, whereas over a longer search horizon, cumulative income effects become more important. This distinction is also consistent with the divergent conclusions in the literature: the work emphasizing the substitution effect studies a single application or a single experiment, whereas the work in which the income effect dominates studies job search over a certain period of time.⁹ The second reason is our population of interest: immigrants. The income effect is generally stronger for “poor” agents, which in the job search context means job seekers who are far from the target they need to reach, and immigrants are more likely to fall in this region. We therefore expect the income effect to be the dominant force in our setting.

2.3 Job Search Behavior

As previously mentioned, job search behavior, Y_m , was measured five months after the intervention as part of wave 3 (section “C. Job Search Behavior” at [this link](#)). In section 3.4, we categorize these questions into seven outcomes, indexed by m . The main questions were framed as follows:

Q1. Since January 2025, have you done any of the following? Please select all that apply. 01 Looked at job postings online, ..., 04 Updated or posted your resume (online or elsewhere), ..., 05-07 Talked about career changes or promotions with ...

Q2. Same as above but in terms of time spent on average per day

Q3. Since January 2025, have you done any of the following to actively search for a job? Please select all that apply. 01 Contacted an employer ..., 02 Contacted an employment agency ..., ..., 05 Applied to a job posting, 06 Attended a career fair, ...

Q4. Same as above but in terms of time spent on average per day

Q5. In the last two months, how many potential employers, if any, did you apply to for employment? Please include all applications made in person, online, or

⁸This argument is the job-search analogue of the income effect in the static labor supply model, and it does not require non-standard preferences. A stronger version of the argument invokes reference-dependent preferences, under which search effort responds to the distance between current and target outcomes rather than to the marginal return alone; see DellaVigna et al. (2017) for evidence from job search, and Camerer et al. (1997) and Fehr and Goette (2007) for the literature on income targeting in labor supply.

⁹Abebe et al. (2025) find that a low-cost job fair corrects overoptimistic beliefs among job seekers in Ethiopia, and that treated job seekers raise their search effort, becoming more likely to visit the job vacancy board within six weeks.

through other direct methods. If you cannot recall the exact number, please make your best guess.

Q6. *In the last two months, how many job interviews did you have, including any informal interviews? If you cannot recall the exact number, please make your best guess.*

Q7. *In the last two months, how many job offers have you received? Please include all offers, including those made informally and those made by contacts not solicited by you. If you cannot recall the exact number, please make your best guess.*

The seven outcomes were chosen to trace successive phases of the job search process. Questions 1 and 2 measure low-cost *preparatory* search such as browsing job postings, updating or posting a resume, and discussing career changes or promotions, where Q1 records whether the respondent undertook each activity (the extensive margin) and Q2 records the time devoted to it (the intensive margin). Questions 3 and 4 capture higher-cost *active* search activities, defined by direct contact with the labor market. The same extensive – intensive pairing applies. Questions 5 through 7 measure *realized* search as counts rather than indicators, including applications submitted, interviews held, and offers received. The two blocks differ in their recall windows: Q1–Q4 are cumulative since January 2025 after the survey and information provision experiment, whereas Q5–Q7 refer only to the two months preceding the survey. The longer window on the first block covers the full post-intervention period and so admits any responses that arrive after the intervention; the shorter window on the second block is designed to limit recall error, which is more severe for Q5–Q7 because those items ask for counts rather than indicators.

3 Results

3.1 Perceived Discrimination

We start by documenting immigrants’ perceived discrimination with respect to X_j^{pre} , that is, immigrants’ pre-intervention beliefs about how they would compare to an otherwise identical native-born worker in the current workplace and during job search.

For current workplace discrimination, panel A in Table 1 shows that on average, first-generation workers believe that a counterfactual native-born worker would generally fare much better in the labor market. Specifically, they expect that a native-born worker would (1) earn 25 percent more in their current wage, (2) work 0.59 more hours, (3) be 7.86 percentage points more likely to be promoted in their current job, and (4) earn 20 percent more in five years.

Turning to job search discrimination, first-generation workers believe that a counterfactual native-born worker would (1) need to send 2.22 fewer CVs to get a callback for an interview and (2) receive job offers with 8 percent higher wages. All of these perceived gaps are statistically different from zero at the 1 percent level, except for current hours worked. Beyond the average belief, all medians are greater than or equal to zero. In particular, the median for the current wage gap of first-generation respondents is 10 percent, and 62.3 percent of individuals believe they face negative wage discrimination (i.e. $\Pr(X_1^{pre} > 0) = 62.3\%$).

Panel B of Table 1 shows the perceived discrimination for second-generation workers. All of their perceived gaps are against themselves and are statistically different from zero at the 1 percent level, except for current hours worked and promotion probability. The last column shows the two-sample t-test of equal means between first- and second-generation respondents. The main takeaway is that second-generation respondents perceive statistically smaller gaps than first-generation immigrants on all outcomes except for current hours.

XX AM: DEPENDING ON HOW MUCH SPACE WE HAVE IT MAY BE GOOD TO INCLUDE THIS FIGURE IN THE PAPER [ZM: Unfortunately, we do not have enough space. Short papers only allow at most 5 exhibitions (including tables and figures), and Angelino and I feel that the current 5 exhibitions in the main text are the most valuable. But let us know if you have different opinions.] Figure 5 in the online appendix shows how perceived discrimination varies with demographic characteristics other than first- or second-generation status. Four main patterns emerge. First, perceived current wage gaps decline with age, so that younger workers report the largest gaps. Second, female workers report higher perceived gaps than male workers. Third, perceived gaps decline with education, so that less-educated workers report the largest wage gaps. Fourth, perceived gaps are larger among workers living outside metropolitan areas. These patterns hold regardless of immigrant generation or wave of the survey (1 or 2).

Table 1: Perceived Discrimination Across First- and Second-Generation Workers

Panel A: First Generation					
	N	Mean	Std. Dev.	Median	p-value (FG = SG)
X_1^{pre} : Current Wage	281	0.246***	0.479	0.100	0.044**
X_2^{pre} : Current Hours	292	0.589	7.928	0.000	0.892
X_3^{pre} : Promotion Probability	266	7.857***	29.214	0.000	0.003***
X_4^{pre} : Future Wage	285	0.204***	0.396	0.055	0.004***
X_5^{pre} : Job Arrival (CV Sent)	284	-2.215***	6.031	0.000	0.008***
X_6^{pre} : Job Search Wage	285	0.083***	0.230	0.000	0.027**
Panel B: Second Generation					
	N	Mean	Std. Dev.	Median	
X_1^{pre} : Current Wage	550	0.174***	0.485	0.042	
X_2^{pre} : Current Hours	562	0.664**	7.436	0.000	
X_3^{pre} : Promotion Probability	490	1.959*	24.168	0.000	
X_4^{pre} : Future Wage	547	0.126***	0.346	0.000	
X_5^{pre} : Job Arrival (CV Sent)	554	-1.162***	5.028	0.000	
X_6^{pre} : Job Search Wage	547	0.048***	0.207	0.000	

Notes: This table presents summary statistics for perceived discrimination gaps across different labor market outcomes. Gaps are calculated as the difference between perceived outcomes for minorities versus non-minorities, and are winsorized at the 5th and 95th percentiles. Stars indicate statistical significance from t-tests that the mean equals zero: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The last column reports the p-value of a two-sample t-test of equal means between first- and second-generation respondents.

3.2 Pessimistic Bias

We now turn to immigrants' perceived discrimination with respect to Z_k at baseline, that is, immigrants' beliefs about how earnings of native-born workers (Z_0) compare with earnings of workers in their respective immigrant group (Z_1 or Z_2) at the population level. On average, first-generation workers believe that native-born workers in their occupation earn 20 percent more than workers in their immigrant group. Second-generation workers believe that native-born workers in their occupation earn 6 percent more than workers in their immigrant group.

To assess whether these expectations reflect any bias, we look at the difference between this perceived discrimination (gap) (Z^{pre}) and the true gap (Z^*) revealed in our information provision experiment based on Census data. As reported in column 4 of Table 2, first-generation immigrants overestimate the true gap by 14 percentage points. Second-generation immigrants also tend to overestimate the magnitude of the gap, even though the difference (5 percentage points) is smaller.

Table 2: Average Gap Between Immigrant Belief and Truth About Annual Earnings

	(1) Native Earning Z_0	(2) Immigrant Earning Z_1 or Z_2	(3) Gap Z^{pre}	(4) Error $Z^* - Z^{pre}$	(5) Abs Error
First Generation	91802 (82357)	80792 (65198)	0.20 (0.64)	-0.14*** (0.23)	0.28 (0.62)
Second Generation	81321 (74540)	79616 (72430)	0.06 (0.69)	-0.05*** (0.17)	0.18 (0.67)

Notes: Column (3) reports the proportional gap, $[(1) - (2)] / (2)$. Column (4) reports the gap relative to the truth, i.e. $\text{Truth} - (3)$. Column (5) reports the mean of $|Z^{pre}|$, the absolute value of Column (4). The outcomes in Column (3) and (4) are winsorized at the 5th and 95th percentiles. Standard deviations are reported in parentheses. A two-sample t -test of the difference in mean error (Column (4)) between first- and second-generation workers yields $p < 0.001$.

3.3 How Does Belief Updating Relate to Bias?

Since we elicited beliefs, X_j , both before and after the intervention, we can examine whether a light-touch, population-level signal shifts individuals' beliefs about the discrimination they themselves face. In particular, we evaluate the relationship between belief updating and the initial bias in perceived population-level wage gaps by estimating the following regression:

$$\underbrace{(X_j^{post} - X_j^{pre})}_{\text{belief revision}} = \phi_{0j} + \phi_{1j} \underbrace{(Z^* - Z^{pre})}_{\text{perceived earnings error}} + C\phi_{cj} + \epsilon_j, \quad (1)$$

where X_j^{post} and X_j^{pre} denote posterior and prior perceived discrimination for measure j , indexing the six variables reported in Table 1. The dependent variable $(X_j^{post} - X_j^{pre})$ thus captures the revision in beliefs from prior to posterior. The term $(Z^* - Z^{pre})$ represents the respondent's error regarding population-level earnings reported in column 4 of Table 2. The vector C includes demographic controls for age, gender, ethnicity, education, marital status, geographic location, homeownership, household size, household income, survey language, and employment status (also see Table 5 in the online appendix). XX AM: Need to include the full list of demographic controls here

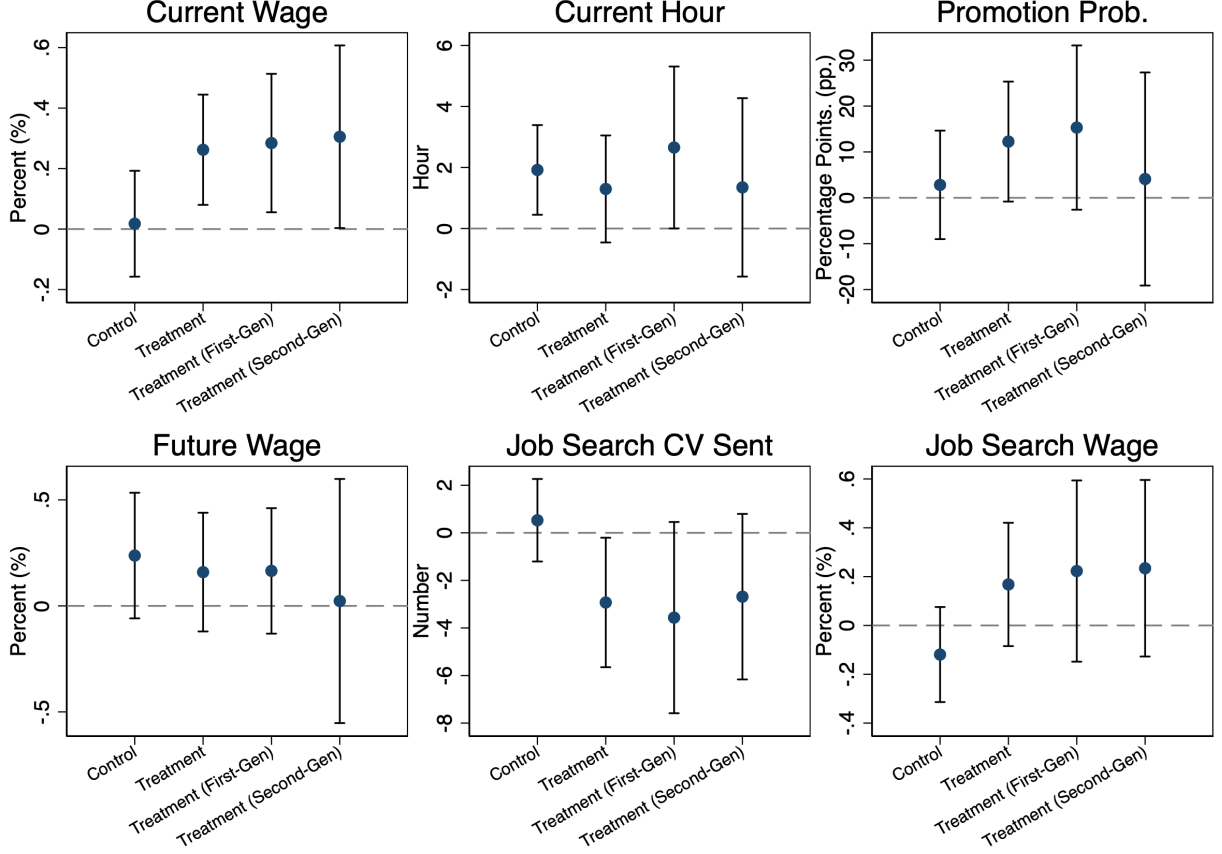
Figure 1 reports the estimated coefficient ϕ_{1j} for the six outcomes discussed in section 2.2 across treatment status and immigrant generation. We restrict the sample to valid-intervention cases in which the occupation category used in the information intervention matches the respondent's current occupation in wave 2.¹⁰ The results indicate that

¹⁰Technical constraints in the survey instrument only allowed us to base the occupation-specific population earnings information on respondents' occupations as reported in the previous wave (2020), rather than on the latest occupation held at the time of the survey and experiment. Because respondents may change occupations between waves, the occupation used in the information intervention does not necessarily coincide

population-level information shifts individual beliefs, as evidenced by the positive and significant ϕ_{1j} for the treatment group, i.e. the second bar for each panel. This suggests that, due to treatment, greater initial overestimation is associated with greater downward revision of perceived discrimination. Stated differently, the more negative and pessimistic one’s initial beliefs on the population-level earnings, the greater the adjustment to one’s priors about perceived discrimination. For example, the third bar of the first panel shows that treated first-generation immigrants who overestimate the population earnings gap by 10 percentage points relative to the true statistic revise their perceived discrimination in current wages downward by 2.8 percentage points, i.e. 11.4 percent of the average prior mean gap, or 0.06 standard deviations of the prior belief gap. The first bar reports the same coefficient for the control group, which is not statistically distinguishable from zero. This indicates that belief revisions for the control group are not associated with the error in population-level beliefs, as these respondents do not receive the signal. The fourth bar shows that the magnitude is similar for second-generation respondents. Note that this is also the case for the “CV Sent” outcome; however, the effect is in the opposite direction because a positive gap indicates greater discrimination against immigrants. Specifically, in the third bar of the fifth panel, among immigrants, overestimating the population earnings gap by 10 percentage points is associated with a downward revision in perceived discrimination of 0.36 CVs sent to obtain a callback, approximately 16 percent of the mean prior gap and 0.06 standard deviations of its distribution. Overall, a key takeaway is that information shifts perceptions of discrimination in both the current workplace (panels 1–4) and the job search process (panels 5–6). Interestingly, the treatment effect is relatively similar across first- and second-generation (i.e. the third and the fourth bar for each panel) though the latter tends to be noisier due to the smaller sample size. XXAM: EXPAND THIS PARAGRAPH A BIT TO MENTION SOME OF THE MAGNITUDES OF THE TREATMENT EFFECTS

with the respondent’s current occupation at the time of the survey. Approximately 31% of respondents changed occupations since the previous wave. XXAM: CLARIFY THE TECHNICAL CONSTRAINTS WE ARE REFERRING TO, OR ALTERNATIVELY DROP THIS FOOTNOTE

Figure 1: Perceived Discrimination Change and Population Belief Gap



Notes: The figure reports estimates of parameter ϕ_{1j} in equation (1), along with 95% confidence intervals based on robust standard errors, for various outcomes and respondent groups. Each panel corresponds to one of the six measures of perceived discrimination, X_j , defined in Table 1 and listed in the same order. Within each panel, the first bar shows the coefficient for the control group, i.e. individuals not shown the Census statistics; the second bar shows the coefficient for the treated group. The third bar shows the coefficient for first-generation respondents in the treated group and the fourth bar shows the coefficient for treated second-generation respondents. The sample is restricted to individuals whose current matches the occupation referenced in the occupation-specific salary information treatment.

3.4 How Does Perceived Discrimination Relate to Job Search?

Finally, we explore the relationship between perceived discrimination and job search. We start with an ordinary least square (OLS) regression, where the dependent variables, Y_m , are the job search activities described in section 2.3 and the independent variables are the second class of beliefs (perceived discrimination) described in section 2.2.

Table 3 reveals some significant associations. XXAM: ARE WE SURE THAT THIS IS THE CORRECT MAGNITUDE? THIS SEEMS SMALL [ZM: I have double-checked our code and results, and I believe the estimates are correct. One reason the coefficient in a single row is relatively small may be that it represents a partial effect—that is, the effect of changing perceived discrimination along one dimension while holding all other dimensions constant.

Because these dimensions are highly correlated, such a coefficient may understate the effect of a broader change in perceived discrimination. Indeed, Panel A of Table 4 shows that a one-standard-deviation increase in the PCA-based composite index of perceived discrimination produces a statistically significant and economically sizable effect—more than half as large as the effect of a 100-percentage-point increase in a single dimension of the perceived wage gap. Perhaps we should reiterate the magnitude of this effect in this section so that readers are not left with the impression that it is economically small. Do you think adding one or two sentences explaining this partial-effect interpretation would be sufficient?] Focusing on X_1^{post} , a 100-percentage-point increase in the perceived current wage gap predicts an 8.6 percent increase in the likelihood of engaging in preparatory job search activities such as looking at job postings, updating or posting one’s resume, or talking about careers with one’s social network (column 1). This represents an additional 14.0 minutes of preparatory job search per day (column 2). In terms of actively searching for a job—for example, contacting employers, attending job fairs, or applying for jobs—this represents a 15.8 percent increase in the likelihood (column 3). This is equivalent to spending an additional 8.6 minutes per day (column 4) and applying to 0.8 more employers (column 5). While these are non trivial magnitudes, these effects do not translate into additional interviews or offers (columns 6-7).

Another gap that is esimated to be significantly associated with job search is the number of resumes/CVs sent to secure a callback, X_5^{post} . A one-unit decrease in the perceived CV-sent gap predicts 1.5 fewer minutes of preparatory job search (column 2) and a 0.9 percent decrease in the likelihood of engaging in preparatory search (column 1). This translates into a 1.3 percent decrease in the likelihood of engaging in active job search (column 3), 0.22 fewer applications submitted (column 5), and 0.03 fewer interviews attended (column 6). Collectively, these findings suggest that higher perceived discrimination predicts higher job search, though this is not reflected across all measures.

Given that the six measures of perceived discrimination, X_j , exhibit effects in the same direction, we construct a composite index, $\bar{X}^{pre}$, based on the first principal component. This approach maximizes statistical power in our two-step estimation. Specifically, we perform principal component analysis (PCA) using the six pre-intervention measures, X_j^{pre} . $\bar{X}^{pre}$ is thus defined as the weighted sum of the standardized variables as follows:¹¹

$$\bar{X}^{pre} = \sum_{j=1}^6 f_j \frac{X_j^{pre}}{sd(X_j^{pre})} \equiv \sum_{j=1}^6 w_j X_j^{pre} \quad (2)$$

Using the same weight w_j , we construct the posterior perceived discrimination index $\bar{X}^{post}$ as follows:

$$\bar{X}^{post} = \sum_{j=1}^6 w_j X_j^{post} \quad (3)$$

To address the potential endogeneity that perceived discrimination is correlated with other factors that also affect job search, we next exploit the randomized treatment in the information provision experiment (see Section 2.1). We instrument $\bar{X}^{post}$ with $T \cdot (Z^* - Z^{pre})$ where T represents the dummy for random treatment assignment in wave 2 and $(Z^* - Z^{pre})$ represents the respondent’s error regarding population-level earnings as discussed in section

¹¹The weights are shown in Table 6 in the online appendix.

Table 3: OLS of Job Search (Wave 3) on Perceived Discrimination (Wave 2)

	(1) Preparatory Search Any	(2) Preparatory Search Minutes	(3) Active Search Any	(4) Active Search Minutes	(5) Employers Applied	(6) Interviews Attended	(7) Offers Received
X_1^{post} : Current Wage Gap	0.0860* (0.0481)	14.00** (6.524)	0.158*** (0.0496)	8.588* (4.794)	0.806** (0.387)	0.102 (0.0871)	0.0564 (0.0821)
X_2^{post} : Current Hour Gap	-0.00208 (0.00331)	-0.716 (0.449)	-0.00644* (0.00341)	-0.262 (0.330)	0.00928 (0.0267)	0.00265 (0.00605)	0.00499 (0.00568)
X_3^{post} : Promotion Gap	0.000986 (0.00123)	0.282* (0.167)	0.00174 (0.00127)	0.122 (0.123)	0.00662 (0.00994)	0.00131 (0.00225)	-0.00194 (0.00212)
X_4^{post} : Future Wage Gap	0.0311 (0.129)	-20.33 (17.51)	0.0218 (0.133)	-3.729 (12.87)	0.235 (1.055)	0.417* (0.238)	-0.0666 (0.225)
X_5^{post} : CV Sent Gap	-0.00897 (0.00559)	-1.568** (0.758)	-0.0127** (0.00576)	-1.005* (0.557)	-0.223*** (0.0452)	-0.0297*** (0.0101)	-0.000637 (0.00960)
X_6^{post} : Job Search Wage Gap	-0.239* (0.142)	-17.16 (19.33)	-0.253* (0.147)	1.599 (14.21)	-0.135 (1.167)	-0.476* (0.263)	-0.000128 (0.248)
Observations	519	519	519	519	506	507	507
Controls	✓	✓	✓	✓	✓	✓	✓

Notes: Preparatory job search activities include: looked at job postings online, in newspapers, magazines, or on bulletin boards; checked union or professional job listings; updated or posted a resume (online or elsewhere); and discussed career changes or promotions with family, friends, coworkers, or people outside of work. Active job search activities include: contacted an employer directly (in person, online, or by email); contacted an employment agency or career center (including university career services); asked friends, family, coworkers, or teachers about job opportunities; reached out to employees at other companies; applied to a job posting (online, in a newspaper, or from other sources); and attended a career or job fair. The continuous outcome variables in columns 2, 4, 5, 6, and 7 are right-winsorized at the 95th percentile. Standard errors in parentheses: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

3.2. The instrument is relevant because, as Figure 1 suggested, the information intervention shifts beliefs. XXAM: NEED TO SAY MORE ABOUT RELEVANCE - IN PARTICULAR STAT. SIGNIFICANCE The last row of Table 4 reports a first-stage F-statistic of 14.6 for the job search activity outcomes in columns 1–4. For the *realized* search outcomes in columns 5–7, the F-statistic slightly falls due to sample size change, but remains above the threshold of 10 in every column. The instrument also meets the exclusion restriction due to randomized treatment. We also control for the composite prior belief, $\bar{X}^{pre}$, and the vector of demographic variables, C , as discussed in section 3.2.

Table 4 presents three panels. Panel A reports OLS regressions as in Table 3; however, the independent variable is the posterior perceived discrimination index, $\bar{X}^{post}$. Consistent with earlier findings, panel A suggests that higher perceived discrimination is positively associated with all measures of job search effort, at both the extensive and the intensive margin. Specifically, individuals whose perceived discrimination is one standard deviation higher apply to one additional job and attend 0.2 additional interviews over the preceding two months. We find no association between perceived discrimination and offers received.

Panel B reports the reduced-form results of regressing outcomes on the instrument $T \cdot (Z^* - Z^{pre})$ and the prior belief $\bar{X}^{pre}$. Although the instrument’s coefficient is statistically insignificant for most job search outcomes, the prior belief of discrimination is strongly and positively correlated with job search intensity, job applications, and interviews attended. This suggests that although the information intervention may not be sufficient to shift individuals’ beliefs and job search outcomes approximately five months later, job search behavior remains strongly affected by prior perceived discrimination.

Finally, panel C reports the two-stage least squares (2SLS) results. While the estimates are uniformly positive for all outcomes, they are imprecisely estimated due to the relatively small sample size. We are thus unable to statistically distinguish the effects of perceived discrimination from zero with one exception: A one-standard-deviation increase in the perceived discrimination index, raises the probability of engaging in preparatory job search by 30.2 percentage points (column 1, panel C).

4 Conclusion

This paper examines how first- and second-generation immigrants perceive labor market discrimination and whether these perceptions shape subsequent job search behavior. We find that immigrants perceive sizable disadvantages relative to otherwise identical native-born workers. Providing occupation-specific earnings information leads respondents with more pessimistic priors to revise their perceptions of discrimination, indicating that such beliefs are responsive to new objective information.

Our results further suggest that perceived discrimination is associated with greater job search effort. Rather than withdrawing from the labor market, immigrants who perceive larger disadvantages appear more likely to prepare for and pursue alternative employment opportunities. While the experimental evidence on long-run behavioral responses is imprecise, our estimates point in a similar direction.

Overall, our findings highlight the importance of beliefs in shaping labor market behavior and suggest that perceived barriers may actually induce increased search effort as workers

Table 4: IV Regression of Job Search (Wave 3) on Perceived Discrimination (Wave 2)

	Preparatory Search Any	Preparatory Search Minutes	Active Search Extensive	Active Search Minutes	Employers Applied	Interviews Attended	Offers Received
Panel A: OLS							
$\bar{X}^{post}$	0.055** (0.024)	7.833** (3.165)	0.085*** (0.024)	8.473*** (2.388)	0.986*** (0.199)	0.161*** (0.048)	-0.012 (0.046)
Observations	351	351	351	351	338	339	340
Panel B: Reduced Form							
$T * (Z^* - Z^{pre})$	0.298* (0.175)	16.432 (23.763)	0.205 (0.177)	9.594 (17.485)	1.448 (1.402)	0.506 (0.345)	-0.021 (0.321)
$\bar{X}^{pre}$	0.060*** (0.019)	5.438** (2.644)	0.074*** (0.020)	5.006** (1.945)	0.718*** (0.162)	0.117*** (0.040)	-0.036 (0.037)
Observations	361	361	361	361	345	346	347
Panel C: IV							
$\bar{X}^{post}$	0.302* (0.180)	12.600 (21.935)	0.175 (0.170)	7.476 (16.227)	1.650 (1.555)	0.557 (0.386)	0.007 (0.351)
$\bar{X}^{pre}$	-0.126 (0.110)	-1.075 (13.442)	-0.038 (0.104)	0.738 (9.944)	-0.264 (0.934)	-0.229 (0.236)	-0.039 (0.213)
Observations	335	335	335	335	323	324	325
Controls	✓	✓	✓	✓	✓	✓	✓
Mean Dep. Var.	0.64	42.76	0.42	25.48	1.85	0.45	0.39
First-stage F-stat	14.60	14.60	14.60	14.60	10.41	10.93	10.72

Notes: The sample is restricted to individuals for whom the occupation used in the wage information treatment matches their actual occupation in wave 2. The continuous outcome variables in columns 2, 4, 5, 6, and 7 are right-winsorized at the 95th percentile. Standard errors in parentheses: *** p<0.01, ** p<0.05, * p<0.10.

attempt to improve their opportunities. Understanding the mechanisms underlying the formation and evolution of these perceptions is fruitful area for future research.

Acknowledgments and AI Use Policy

This research was supported by the National Science Foundation (award 1832144). It was approved by the Spelman College Institutional Review Board (ID 6901D2) and pre-registered as AEARCTR-0013494. We are grateful to Melanie De La Rosa, Hannah Hailemichael, Jadyn Jones, Sarah Lokenauth, and Gabrielle Strawn for research support. Generative AI was used to (1) produce the first draft of the conclusion, (2) check the final draft of the paper for typos and inconsistencies, (3) search for BibTeX citations, and (4) help conduct data analysis in Stata.

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Online Appendix

Figure 2: Sample Size across Study Waves

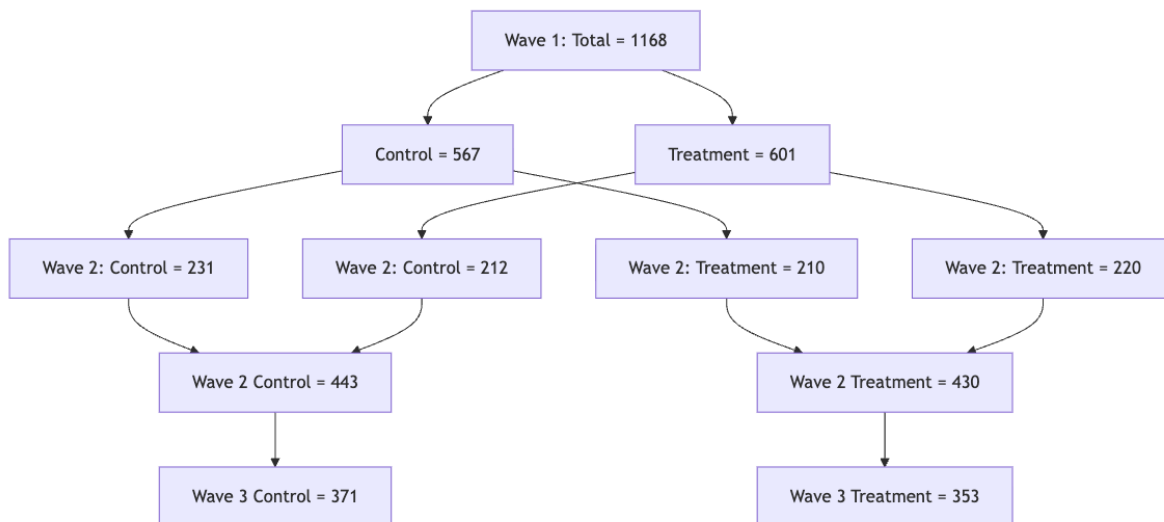


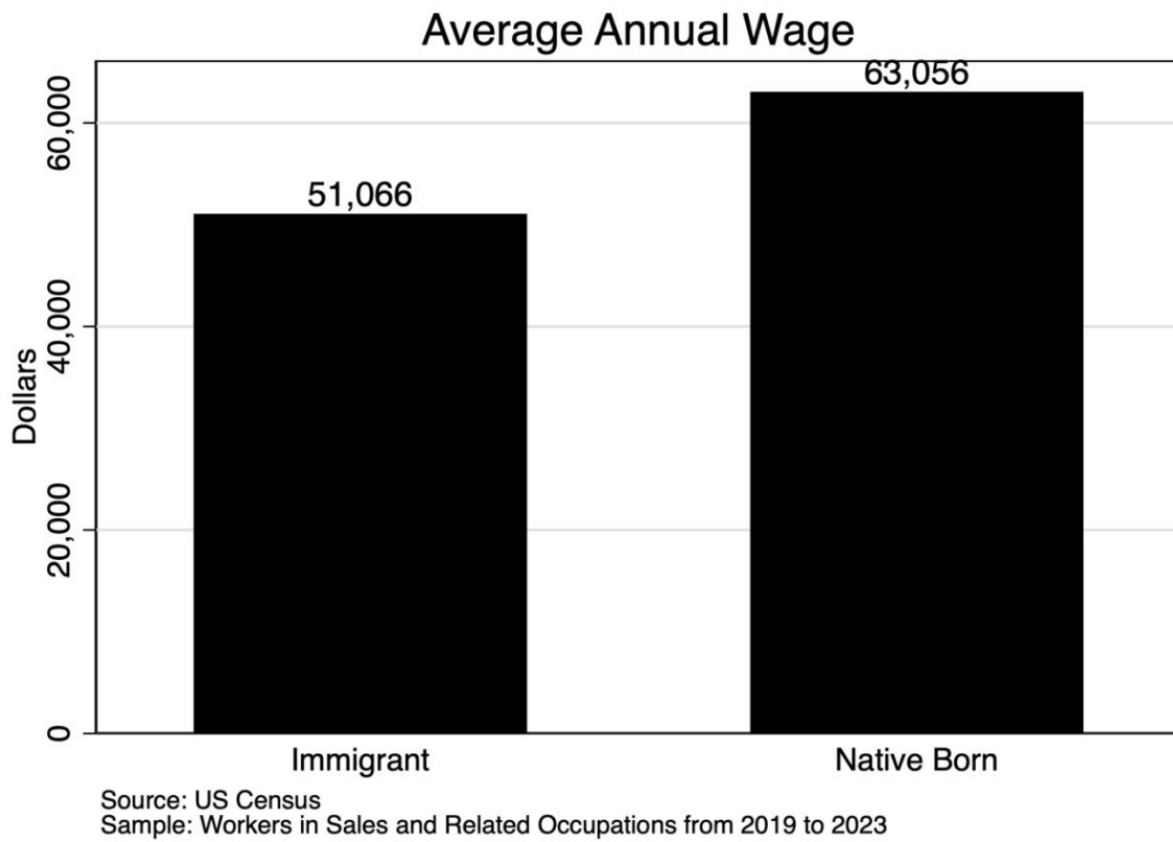
Table 5: Internal Validity – Balance of Observables across Treatment and Control

	N	All	Control	Treated	p-value
<i>Demographics</i>					
Age	873	43.81	44.12	43.50	0.47
Female	873	0.46	0.45	0.47	0.59
Black	873	0.07	0.06	0.07	0.43
Hispanic	873	0.40	0.40	0.39	0.74
Asian	873	0.19	0.20	0.18	0.36
Some College	873	0.28	0.27	0.30	0.34
Completed College or Above	873	0.55	0.55	0.55	0.99
Metropolitan Area	873	0.94	0.93	0.94	0.55
Northeast	873	0.14	0.14	0.14	0.99
Northwest	873	0.18	0.18	0.17	0.61
South	873	0.33	0.34	0.31	0.32
West	873	0.36	0.33	0.38	0.17
Married	873	0.56	0.57	0.55	0.55
Owns Home	873	0.63	0.65	0.61	0.33
Rents Home	873	0.35	0.34	0.36	0.45
Household Size of One	873	0.11	0.11	0.11	0.86
Household Size of Two	873	0.27	0.25	0.28	0.34
Household Size of Three or Above	873	0.62	0.64	0.61	0.45
Household Income Below 30,000	873	0.14	0.14	0.14	0.91
Household Income Below 60,000	873	0.36	0.36	0.37	0.96
Household Income Below 100,000	873	0.62	0.64	0.61	0.33
English-Surveyed	873	0.92	0.92	0.91	0.71
Part-Time Work	837	0.84	0.84	0.83	0.87
Working	869	0.96	0.96	0.96	0.78
Second-Generation Immigrant	873	0.65	0.68	0.63	0.15
<i>Wave 2 Prior Belief Gap Among Respondents and Counterfactual Natives</i>					
Current Wage Gap	831	0.20	0.20	0.20	0.89
Current Working Hour	854	0.64	1.01	0.25	0.14
Promotion Probability	756	4.03	6.20	1.93	0.02**
Future Wage	832	0.15	0.16	0.15	0.58
Resume Sent for Callback	838	-1.52	-1.33	-1.71	0.30
Job Search Wage	832	0.06	0.06	0.06	0.80
Bias in Population Earning Gap	804	-0.08	-0.08	-0.08	0.81
<i>Wave 1 Belief Revisions</i>					
Current Wage	800	0.01	0.01	0.01	0.98
Current Working Hour	836	-0.08	-0.01	-0.15	0.59
Promotion Probability	741	0.77	0.39	1.16	0.57
Future Wage	802	-0.08	-0.07	-0.09	0.32
Resume Sent for Callback	810	0.08	-0.11	0.27	0.08*
Job Search Wage	800	-0.03	-0.02	-0.03	0.78

Table 6: PCA Factor Loadings

	Current Wage	Current Hour	Promotion Prob.	Future Wage	CV Sent	Job Search Wage
Loading	0.448	0.224	0.369	0.532	-0.347	0.458

Figure 3: Example of Treatment Information for First-Generation Respondent

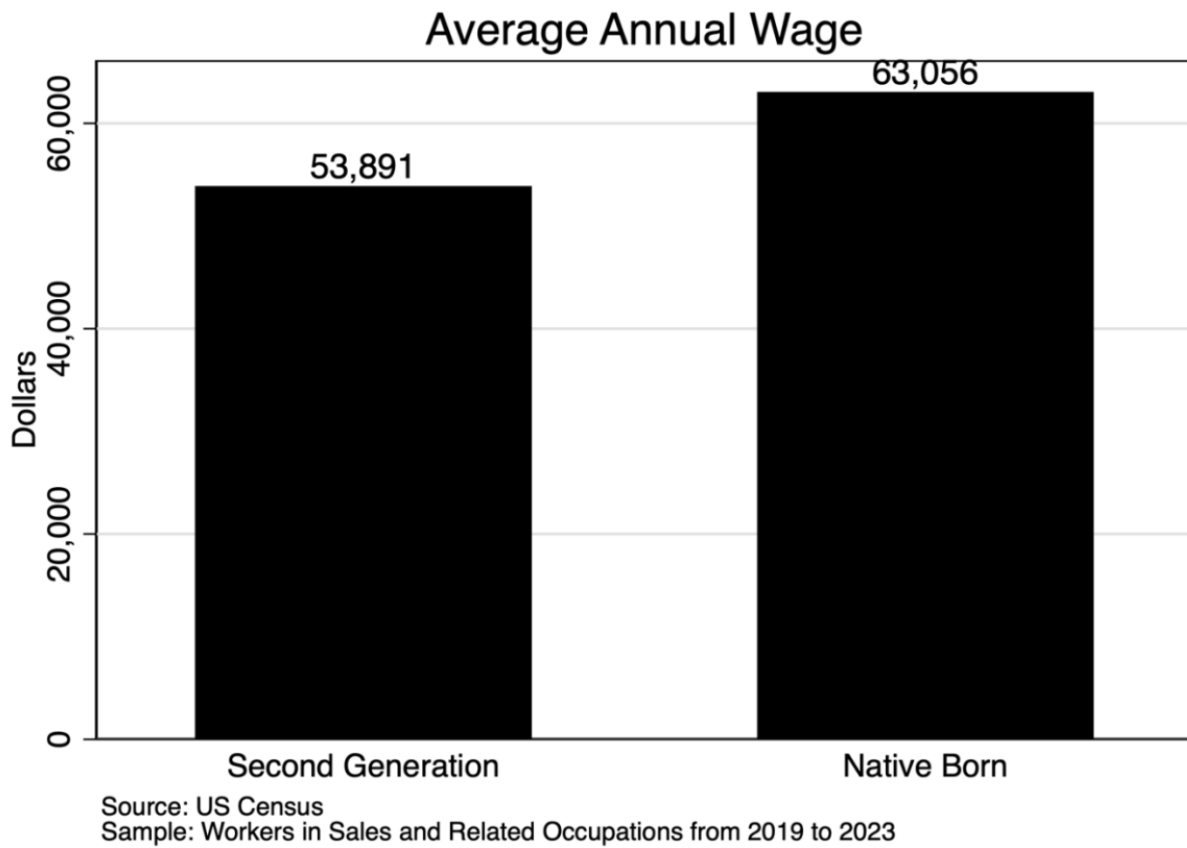


The first bar says that immigrant American workers who have the same job as you on average earned \$51,066 annually.

The second bar says that native-born workers in the U.S. who have the same job as you on average earned \$63,056 annually.

Therefore, immigrant Americans earned \$11,990, or 19% less than native-born Americans on average in the same job as you.

Figure 4: Example of Treatment Information for Second-Generation Respondent



The first bar says that second-generation American workers who have the same job as you on average earned \$53,891 annually.

The second bar says that native-born American workers who have the same job as you on average earned \$63,056 annually.

Therefore, second-generation Americans earned \$9,165 or 15% less than native-born Americans on average in the same job as you.

Figure 5: Perceived Discrimination by Demographics, Immigrant Generation, and Wave

